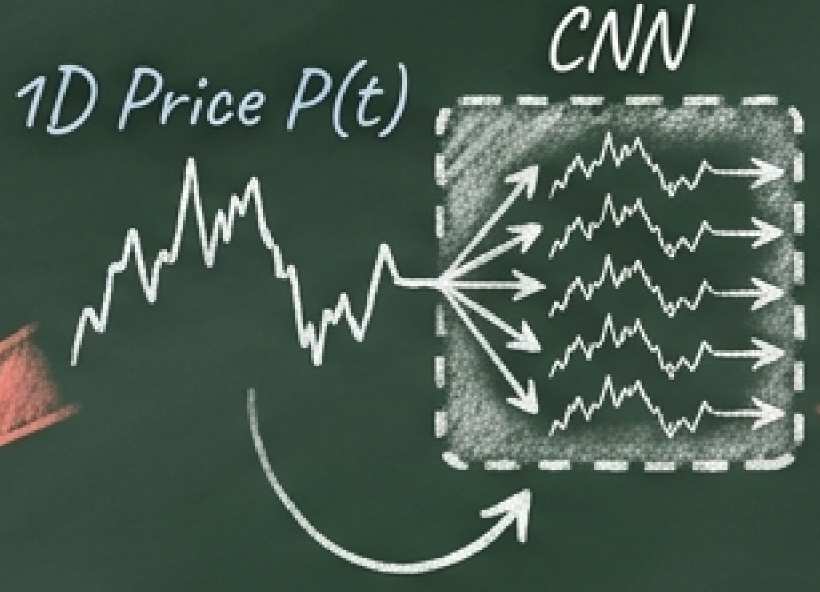
From 1D Price to Multivariate Physics

Upgrading the PhaseSync SDE with VMD & EMD Signal Decomposition

The Flaw of Artificial 1D "Lifting"

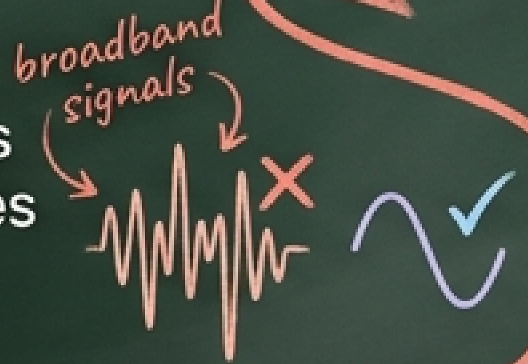
In the original SDE12A03X model, a CNN duplicates a single 1D price into 48 "virtual" oscillators.
This breaks two fundamental laws of physics-informed AI:



1. The Broadband Violation

Daily prices contain mixed frequencies. The Kuramoto model strictly assumes narrow-band oscillators.

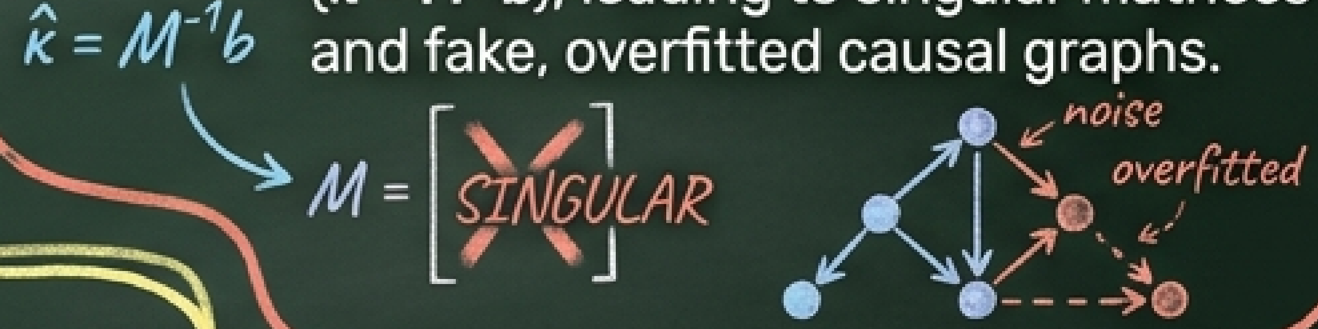
Applying Hilbert transforms to broadband signals causes phase jumps and Gibbs distortion.



2. Spurious Coupling

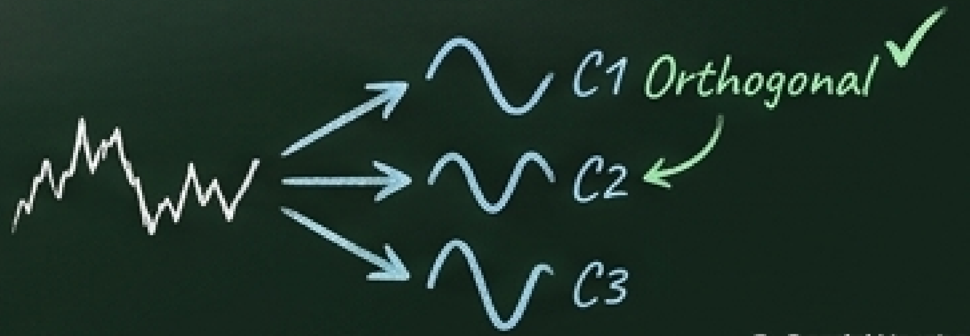
Artificial cloning means all 48 oscillators share 100% correlated noise.

This destroys the "Spatial Conditional Independence" required for the Stanford KvM Closed-form MLE ($\hat{\kappa} = \mathbf{M}^{-1}\mathbf{b}$), leading to singular matrices and fake, overfitted causal graphs.



Solution:

We must decompose the 1D signal into real, orthogonal physical components before it enters the SDE.

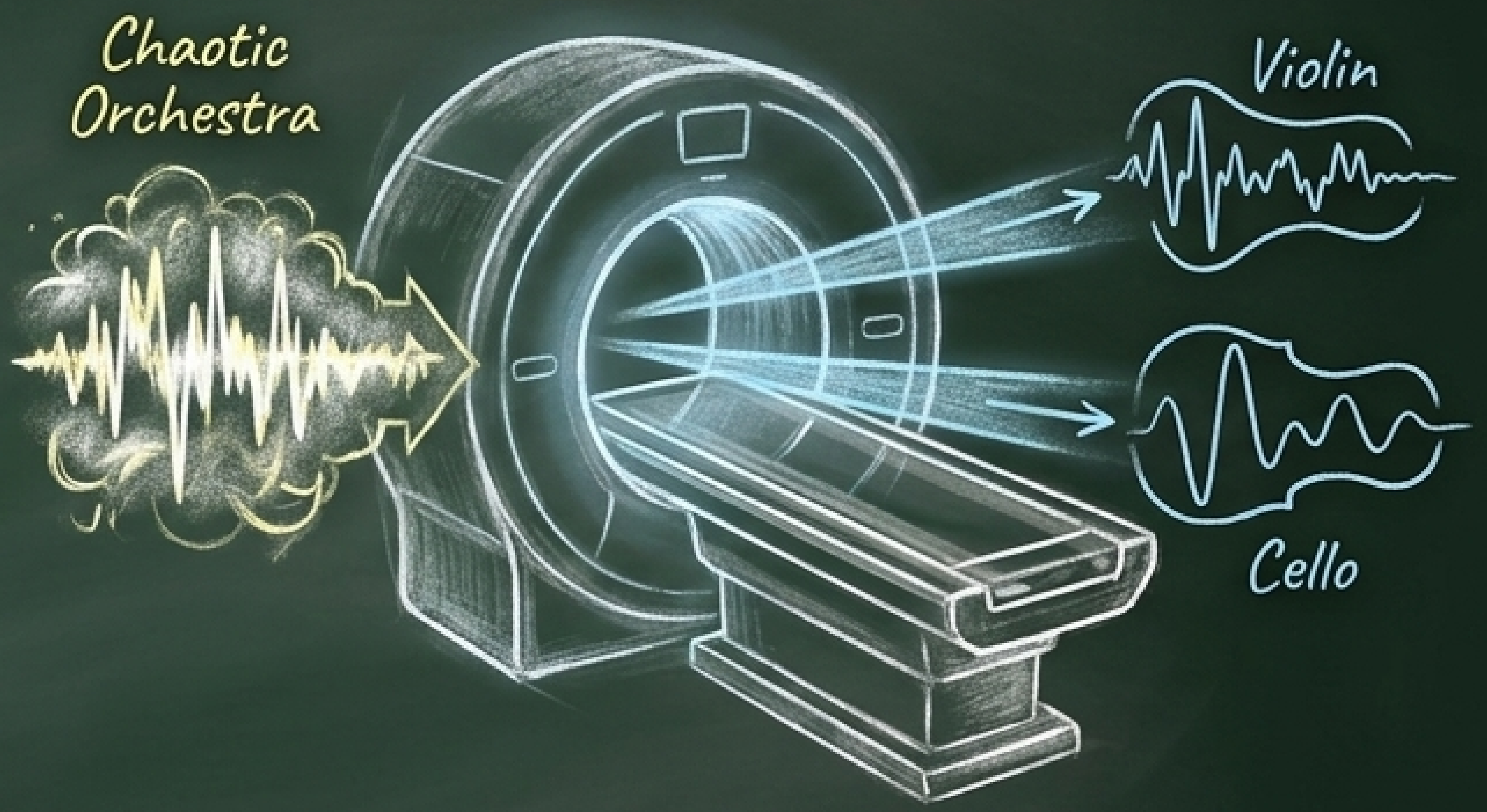


EMD vs. VMD: The Sieve vs. The Scalpel

Empirical Mode Decomposition: The "Sieve" Approach

- Filters data locally through repeated sifting. Like passing a pile of dirt through increasingly fine sieves.
- Highly intuitive, data-driven, free of predefined mathematical rules.

Variational Mode Decomposition: The "Scalpel" Approach

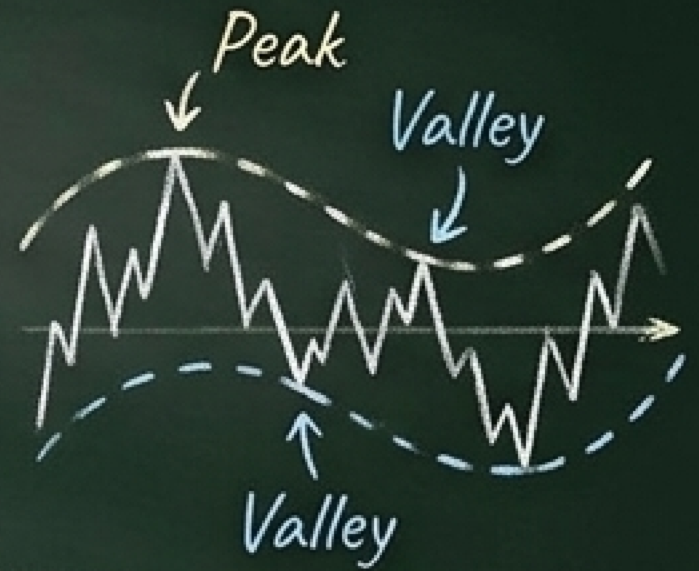


- Global mathematical optimization. Like a scanner declaring, "This is an orchestra of *K* instruments," and separating them simultaneously.
- Mathematically rigorous, concurrent, optimization-driven.

How EMD Works (and its Fatal Flaw)

1. Find Extrema: Locate local peaks and valleys in the data.

2. Draw Envelopes: Connect peaks (upper) and valleys (lower) using cubic splines.

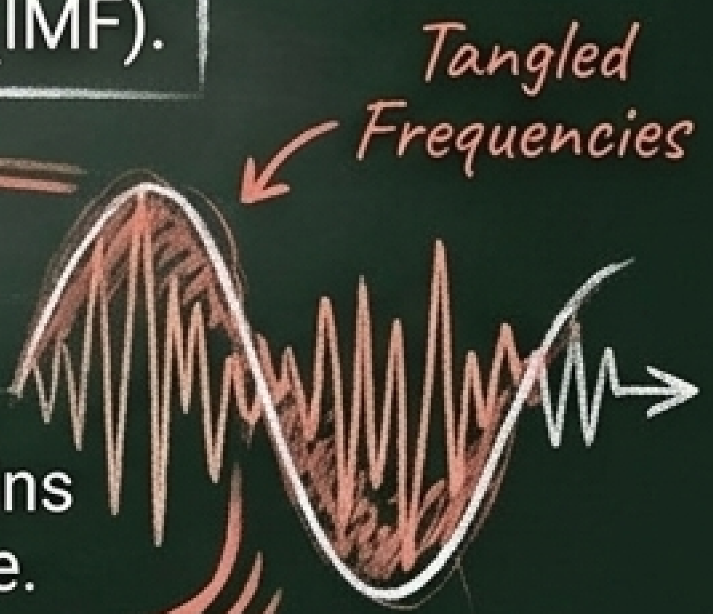


3. Extract Residual: Subtract the mean of the envelopes from the signal.

4. Sift: Repeat until the residual becomes an Intrinsic Mode Function (IMF).

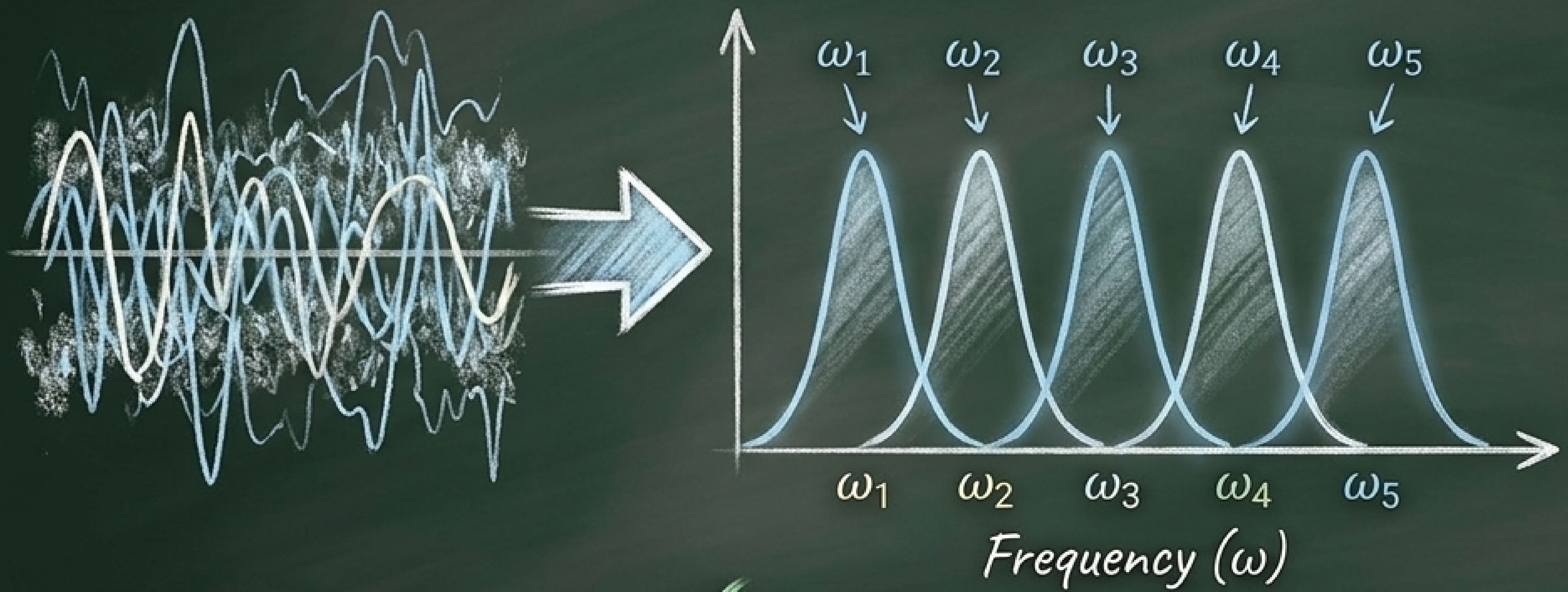
The Fatal Flaw: Mode Mixing

EMD lacks a governing equation. If a sudden news shock hits the stock price, high and low frequencies randomly tangle into the same IMF. This ruins the unique natural frequency (ω_i) required for our Kuramoto physics engine.



How VMD Works (The Mathematical Cure)

VMD abandons step-by-step sifting. Instead, it extracts K modes simultaneously by forcing each mode to pack tightly around a unique center frequency (ω_k).



Key Mechanisms:

- **Analytic Signal**: Uses the Hilbert transform to strip away negative frequencies.
- **Baseband Shift**: Shifts the mode directly to the origin ($e^{-j\omega_k t}$).
- **L2 Minimization**: Minimizes the gradient to force extreme "narrow-band" smoothness.

Result: Pure, distinct frequency bands that are completely immune to EMD's mode mixing.

Selecting the Right Tool for the SDE Engine

Feature	EMD	VMD
Mechanism	Sequential Local Sifting	Simultaneous Global Optimization
Mathematical Basis	Heuristic / Empirical	Variational Constraints (ADMM)
Mode Mixing	Highly Vulnerable ✗	Completely Immune ✓
Frequency Output	Unpredictable Broadband	Perfect Narrow-band ✓
SDE Compatibility	Poor (Breaks ω_i logic)	Perfect (Matches Kuramoto physics)

Verdict: VMD is the required choice to build a mathematically defensible Physics-AI Hybrid model.

Inside the VMD Engine: The Constrained Optimization

Hilbert Transform: Strips away negative frequencies to create an analytic signal.

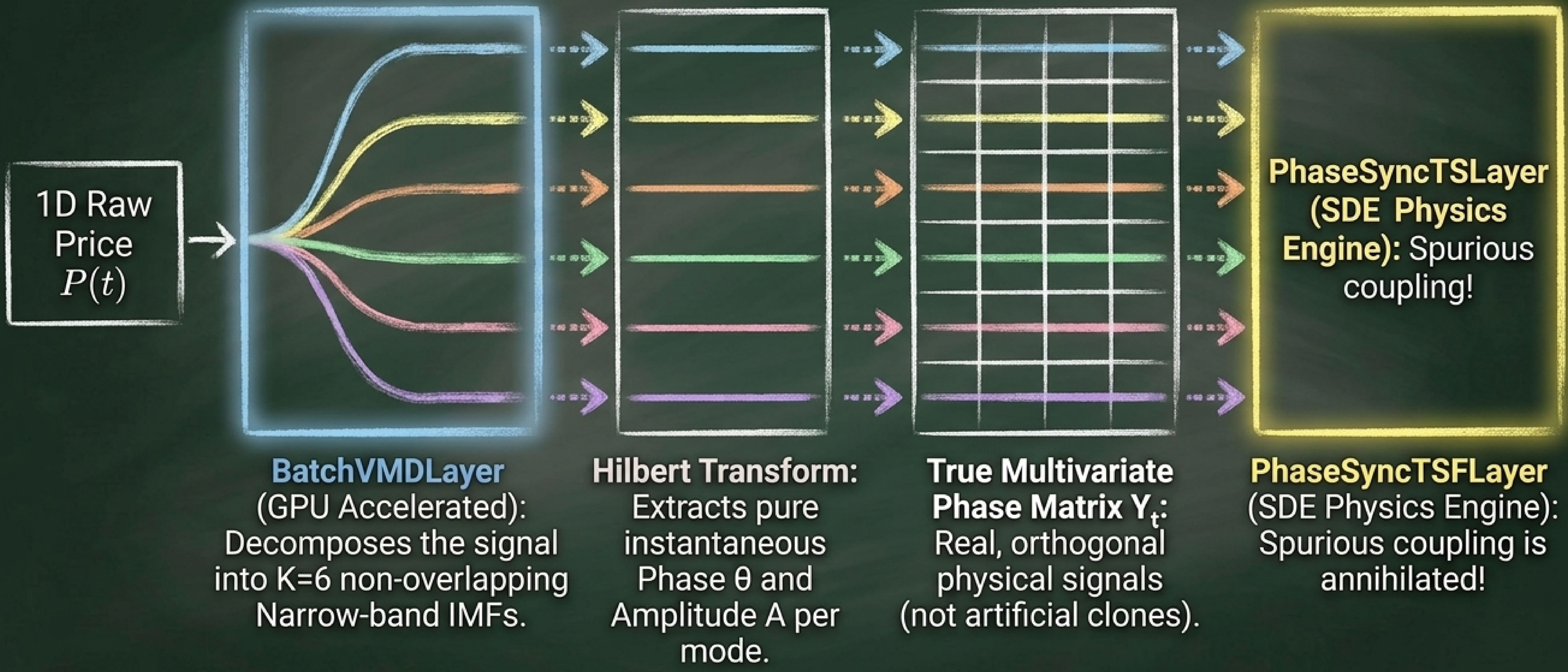
Baseband Shift: Pulls the wave to the center frequency (ω_k) to stabilize it.

$$\min_{u_k,\omega_k} \sum_k \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_k(t) \right] e^{-j\omega_k t} \right\|_2^2 \quad \text{s.t.} \sum_k u_k = f(t)$$

L2 Gradient Minimization: Forces the wave to be perfectly smooth and strictly ‘narrow-band”. No jagged noise allowed!

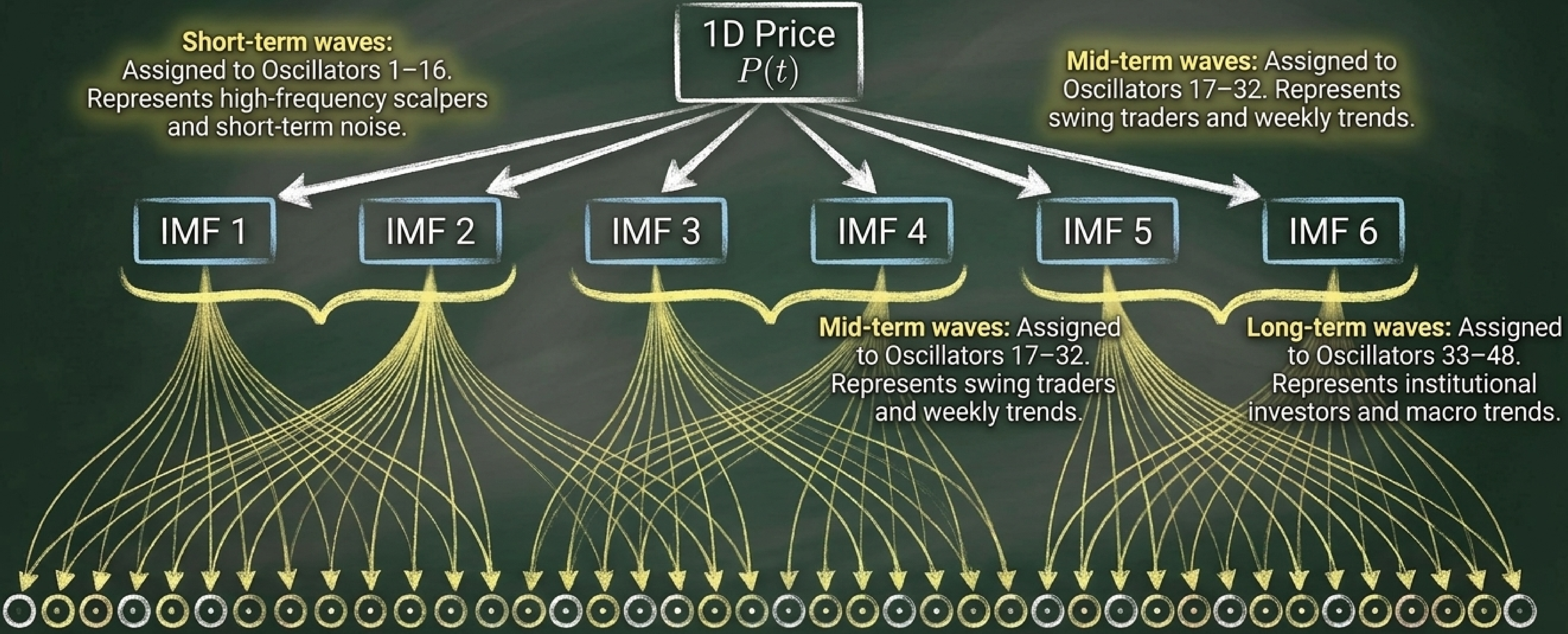
Conservation Law: All separated modes must perfectly sum up to the original raw stock price.

Integrating VMD into PhaseSyncTSF



From 1 Price to 48 Oscillators: The 1:8 Subgroup Mapping

Instead of duplicating the same data 48 times, we distribute **6 distinct**, mathematically **orthogonal physical signals** to represent different market participants.



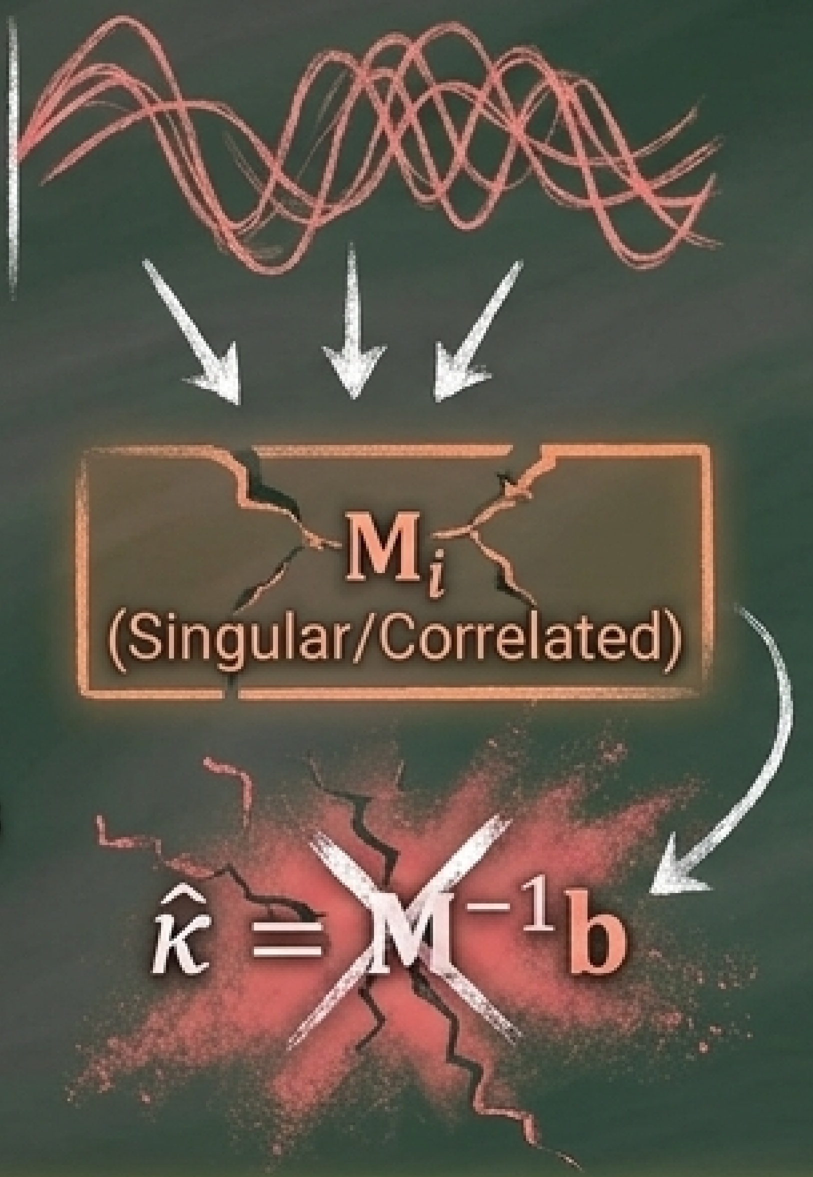
Why This Matters: Saving the Stanford Closed-Form MLE

Before VMD: Failed Spatial Independence

CNN clones meant 100% correlated noise across nodes.

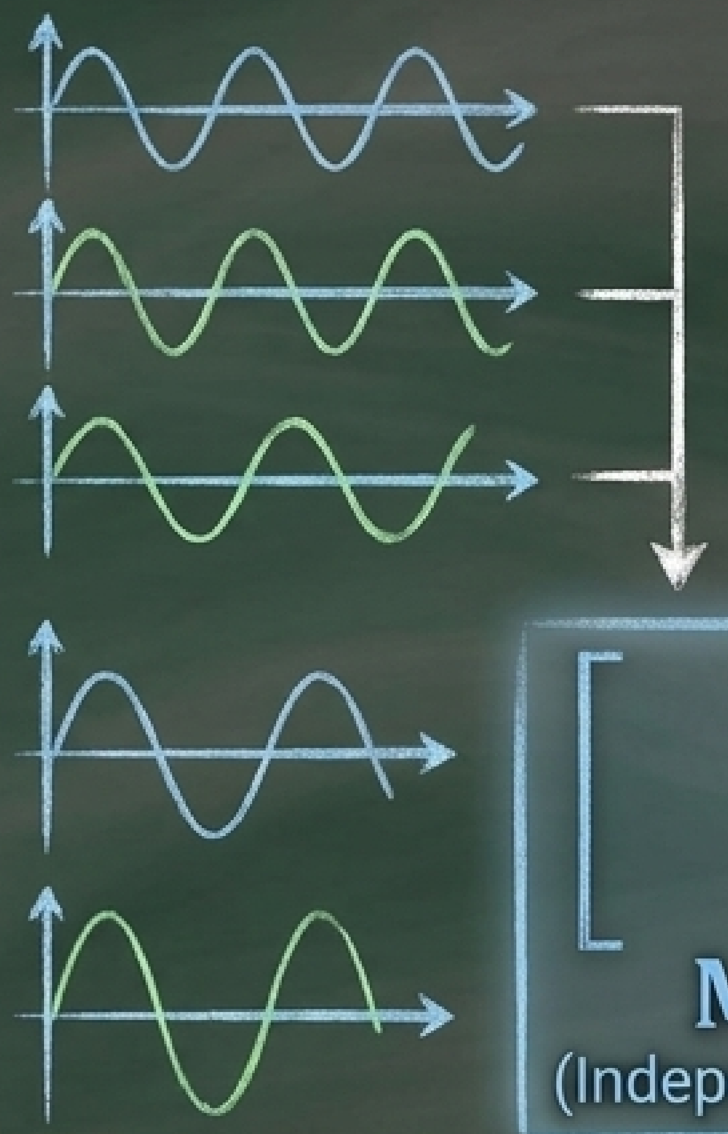
The Covariance Matrix $\mathbf{M}_i$ became singular.

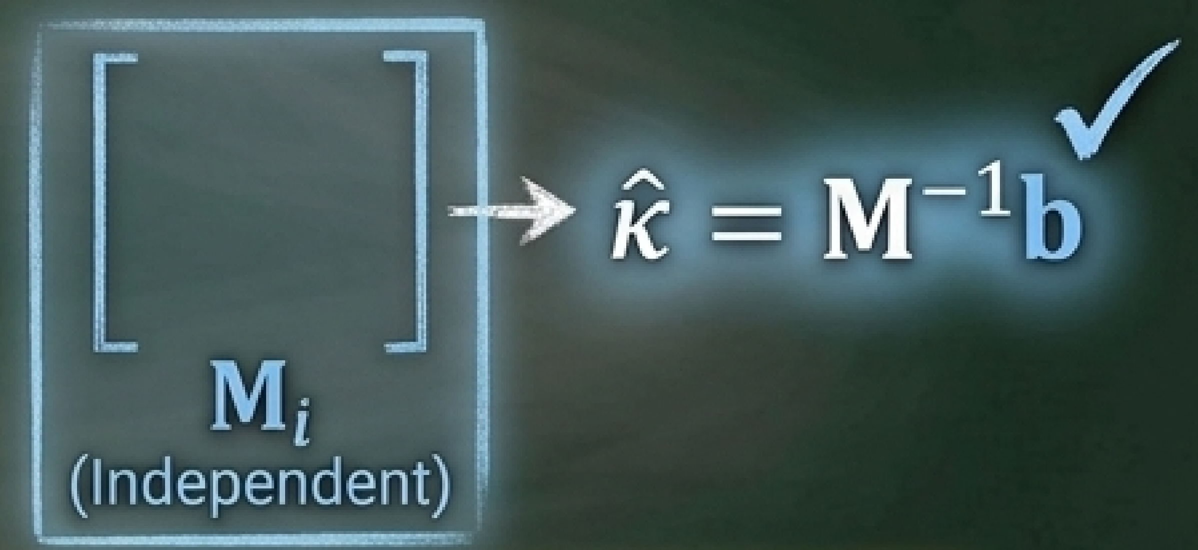
The MLE solver ($\hat{\kappa} = \mathbf{M}^{-1}\mathbf{b}$) panicked and created massive 'Spurious' couplings just to fit the noise.



After VMD: Independence Restored

Because VMD modes are strictly orthogonal in the frequency domain, the noise across subgroups is statistically independent.

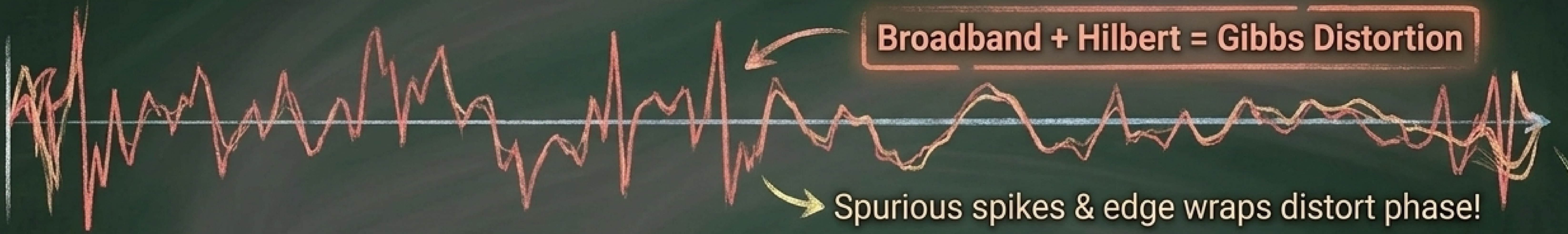




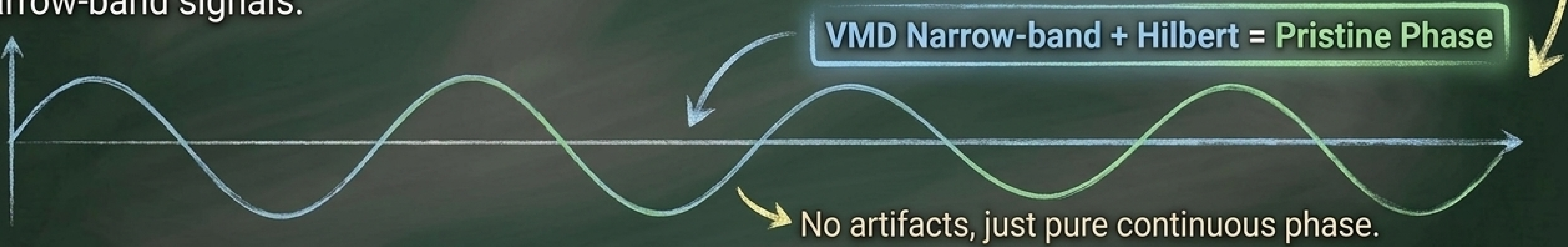
The **Stanford KvM Factorization** holds true! The exact causal Lead-Lag relationships between short-term and long-term market trends are perfectly uncovered.

Eliminating Gibbs Distortion for Clean Phase Extraction

The Problem: Running a Hilbert transform directly on a raw, broadband stock price causes extreme derivative spikes and edge wrapping artifacts (Gibbs effect).



The VMD Fix: Because VMD acts as a perfect geometric scalpel, it outputs strictly narrow-band signals.

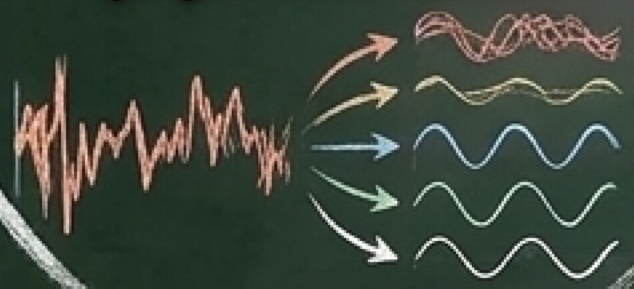


The **Outcome**: Applying Hilbert transforms to these narrow-band IMFs yields a flawless, continuously flowing instantaneous phase $\theta_k(t)$ and amplitude $A_k(t)$ without any mathematical artifacts.

The PhaseSync VMD-SDE Pipeline Complete

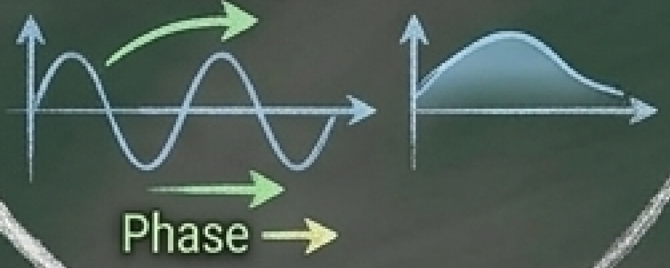
1. Decompose
(VMD)

Raw price is mathematically separated into 6 orthogonal physical realities.



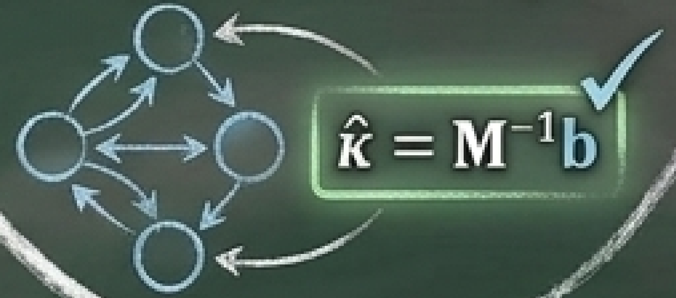
2. Extract
(Hilbert)

Clean phase and amplitude are extracted perfectly perfectly per mode.



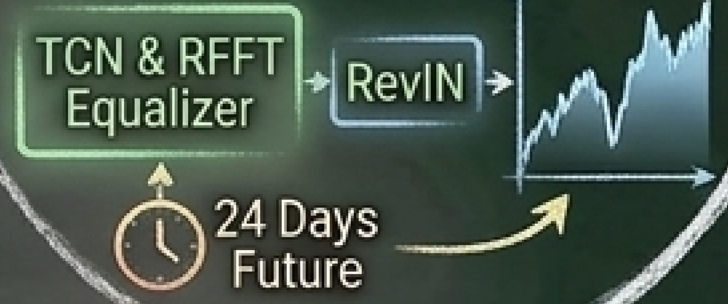
3. Interact
(Kuramoto SDE)

Oscillators interact via a valid, non-spurious asymmetric coupling matrix ($\hat{\kappa} = \mathbf{M}^{-1}\mathbf{b}$).



4. Reconstruct
(TCN & RFFT Equalizer)

Phases are simulated 24 days into the future and remixed with real-world price amplitudes via RevIN.



The **Final Verdict**: By replacing simple CNN lifting with Variational Mode Decomposition, the model evolves from an "engineering shortcut" to a Journal-Grade Physics-AI Hybrid System.